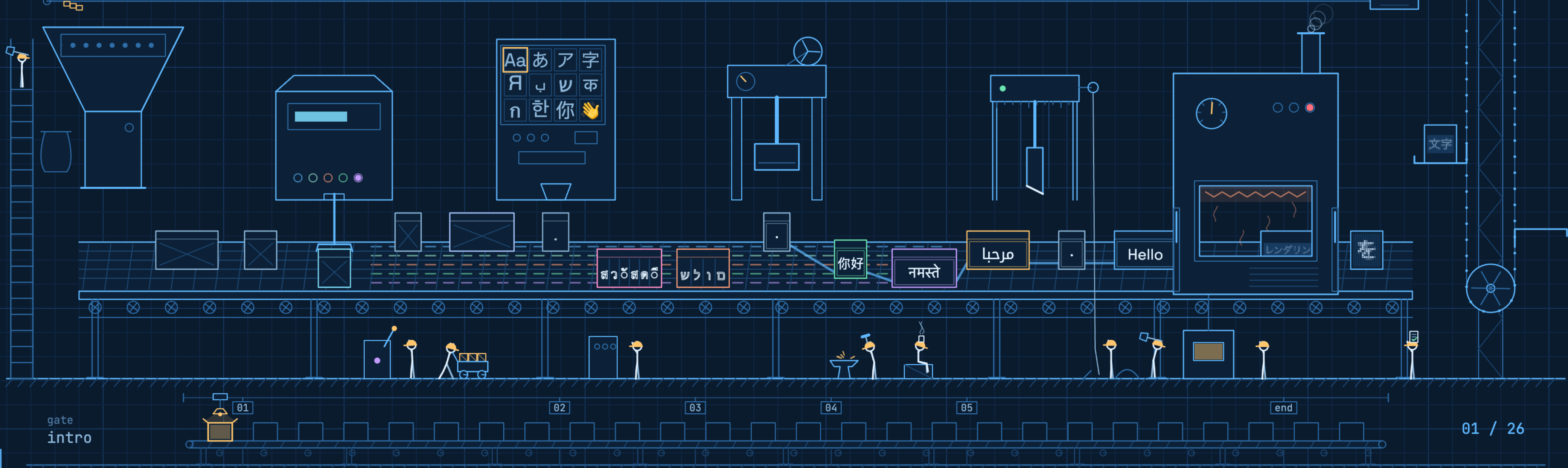


Inside Flutter's Text Pipeline

Flutterテキストパイプラインの内側

で文字をレンダリング・Hello · مرحبا · नमस्ते · 你好 · டுஷ · สวัสดี · Привет · 안녕 · 🖐️ · こんにちは · 日本語 · 縦書き · Fl



01

hall 01 · the line

From code points to pixels

「コードポイントからピクセルへ」

unicode

628 U+0915 U+65

U+0915

U+0628

U+0041

shape

raster

A string ≠ text



Hello が 🇯🇵 👤 नमस्ते

graphemes



s.characters.length 1

code points

 U+1F468	ZWJ U+200D	 U+1F469	ZWJ U+200D	 U+1F467	ZWJ U+200D	 U+1F466
--	---------------	--	---------------	--	---------------	--

7

UTF-16 units

D83D	DC68	200D	D83D	DC69	200D	D83D	DC67	200D	D83D	DC66
------	------	------	------	------	------	------	------	------	------	------

s.length 11

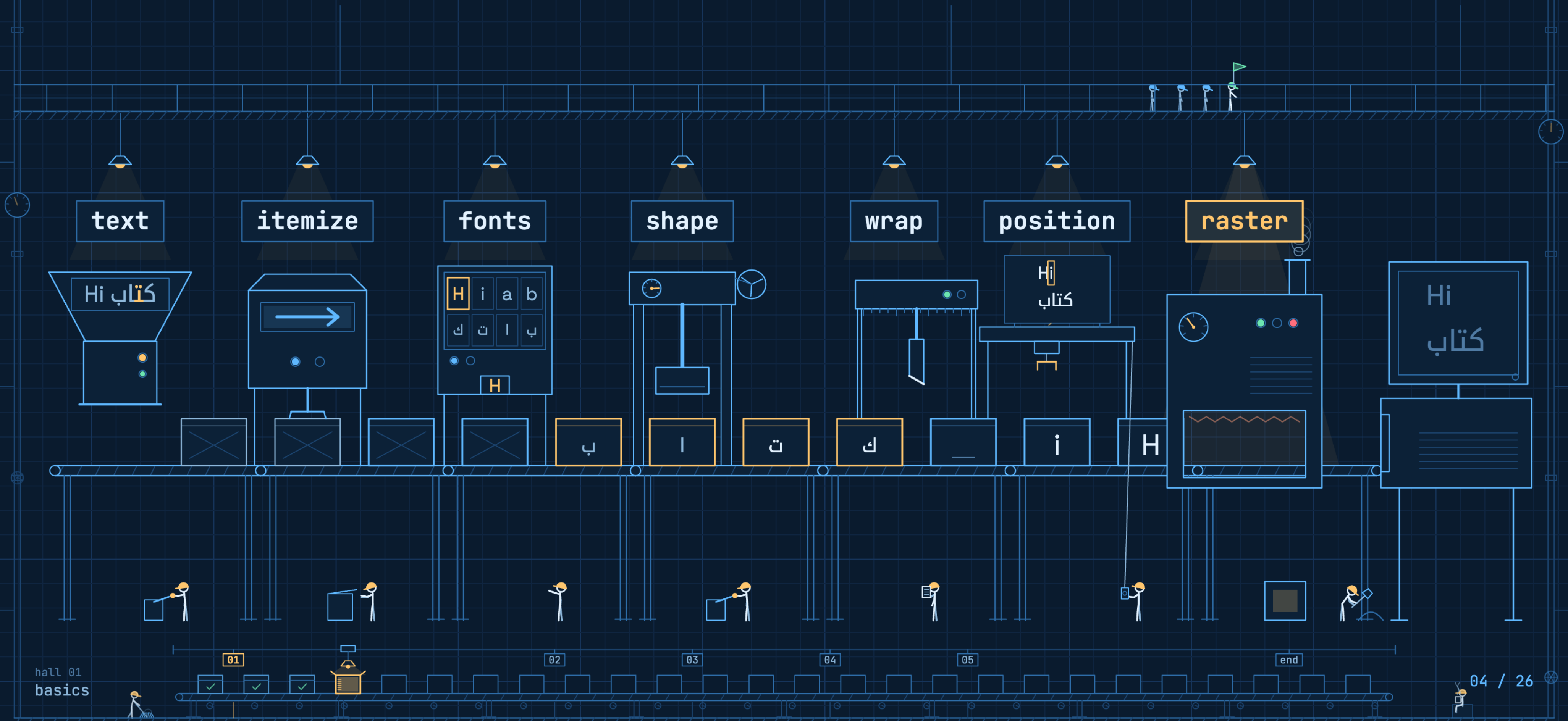
UTF-8 bytes

F0	9F	91	A8	E2	80	8D	F0	9F	91	A9	E2	80	8D	F0	9F	91	A7	E2	80	8D	F0	9F	91	A6
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

25



The pipeline



Itemize

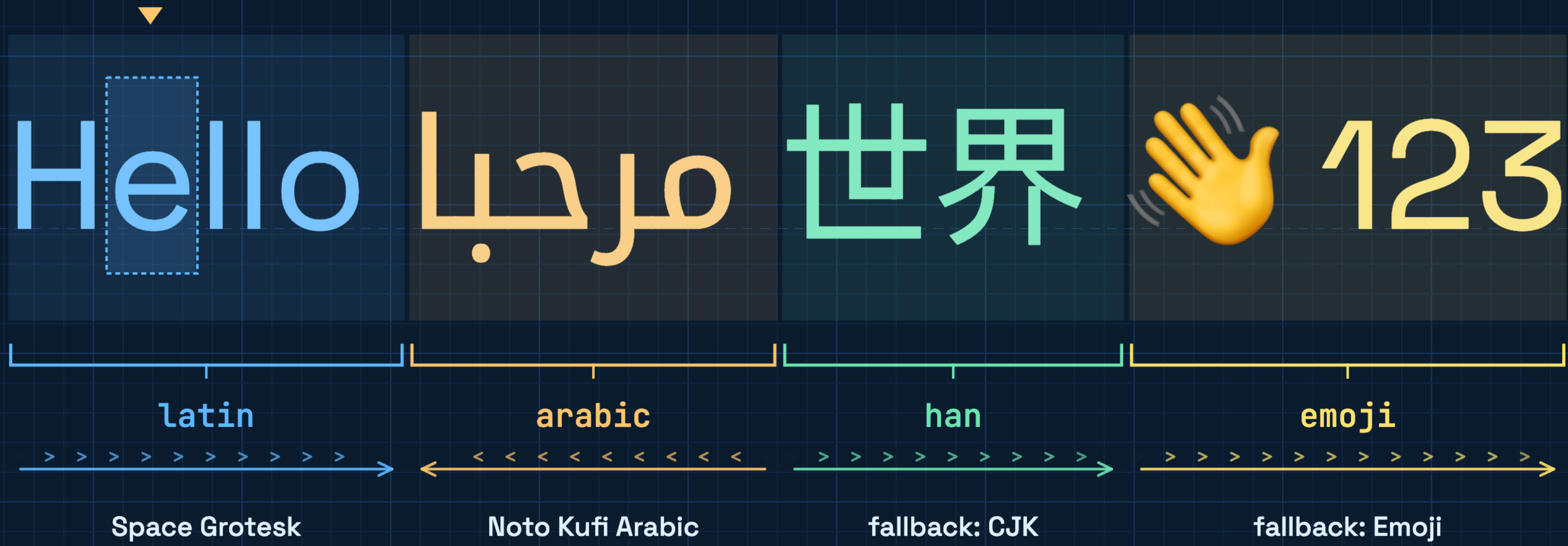
› Hello مرحبا 世界 🖐️ 123

Hello مرحبا 世界 🖐️ 123

Flutter で 日本語 🇯🇵

Hi 2026 کتاب

नमस्ते Привет שלום



Shaping

joining

ligatures

kerning

reordering

emoji ZWJ

←
كتاب

joined

Line breaking

English

Japanese

German

Thai

Arabic

← `maxWidth 520` →

Greedy line breaking
puts as many words on
each line as will fit, then
moves on to the next.

Rasterization

a

g

ξ

あ

字

aliased

grayscale

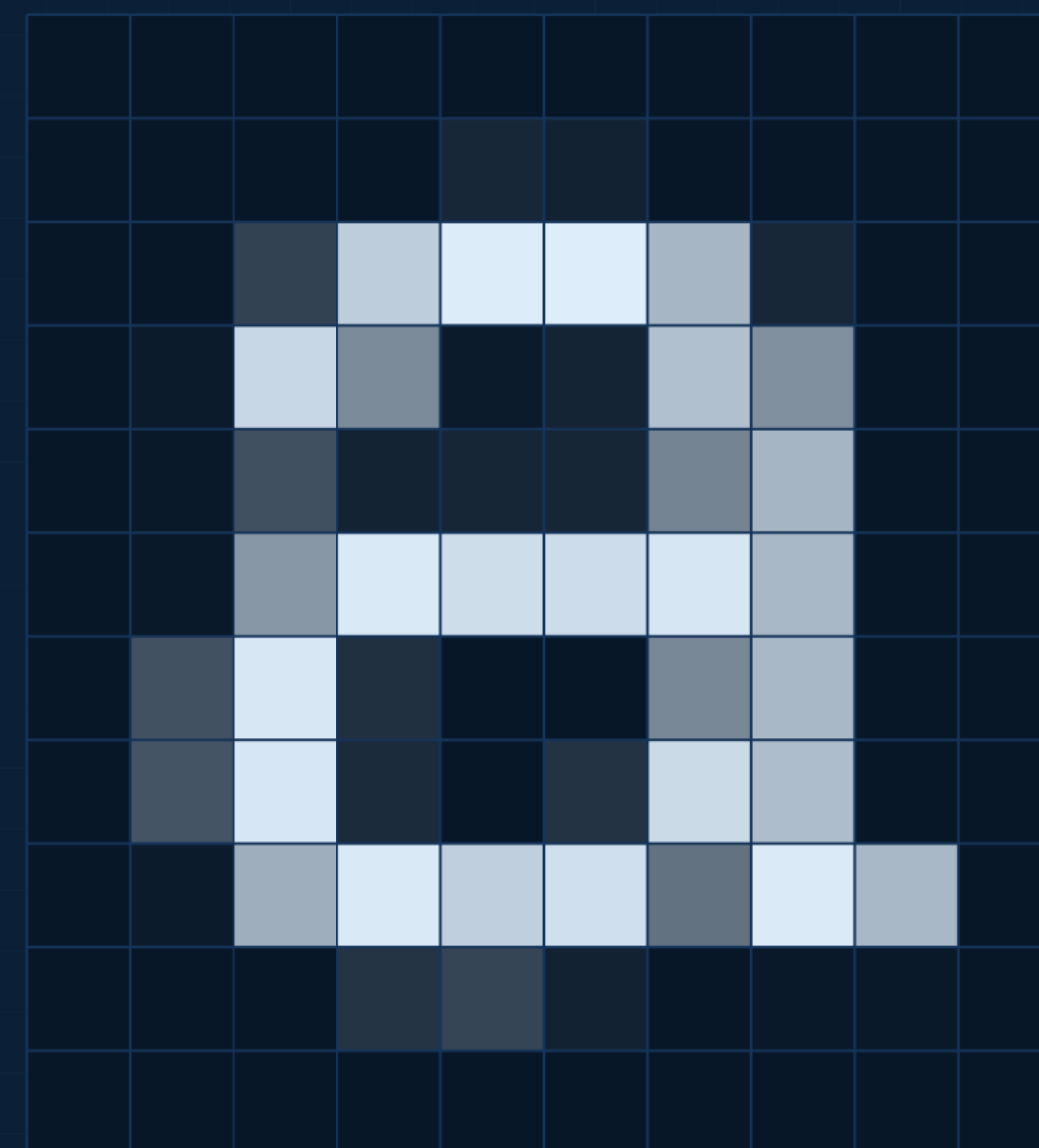
subpixel

14 px

outline



pixels

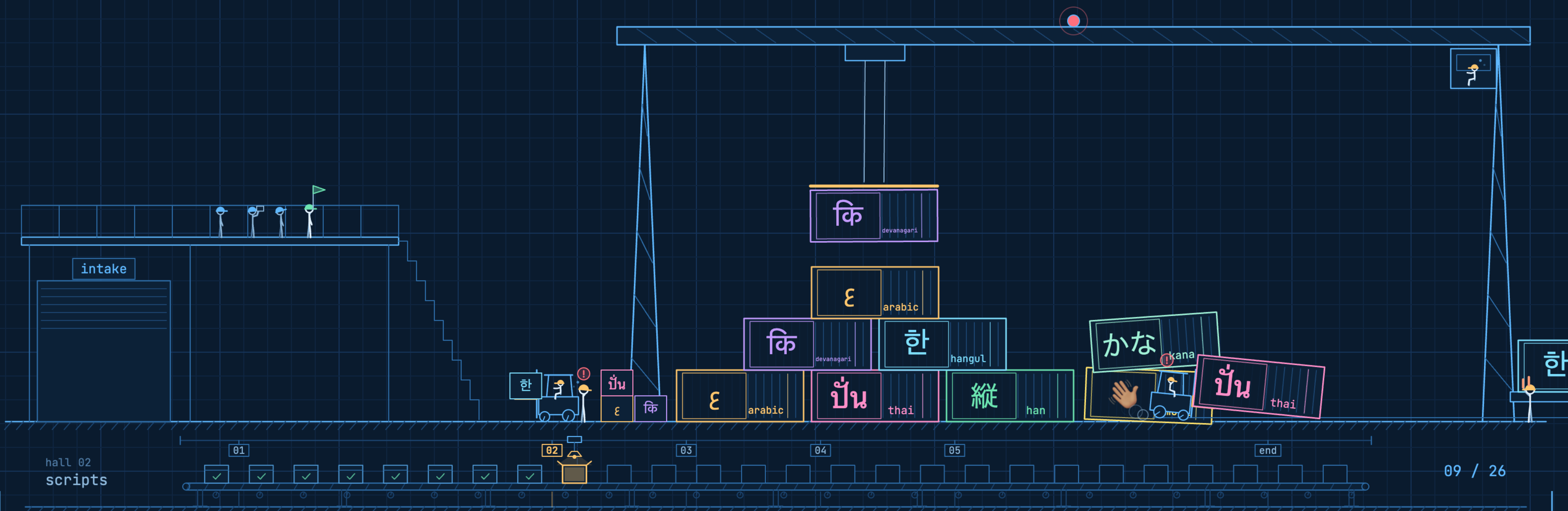


02

hall 02 · imports

Every script adds rules

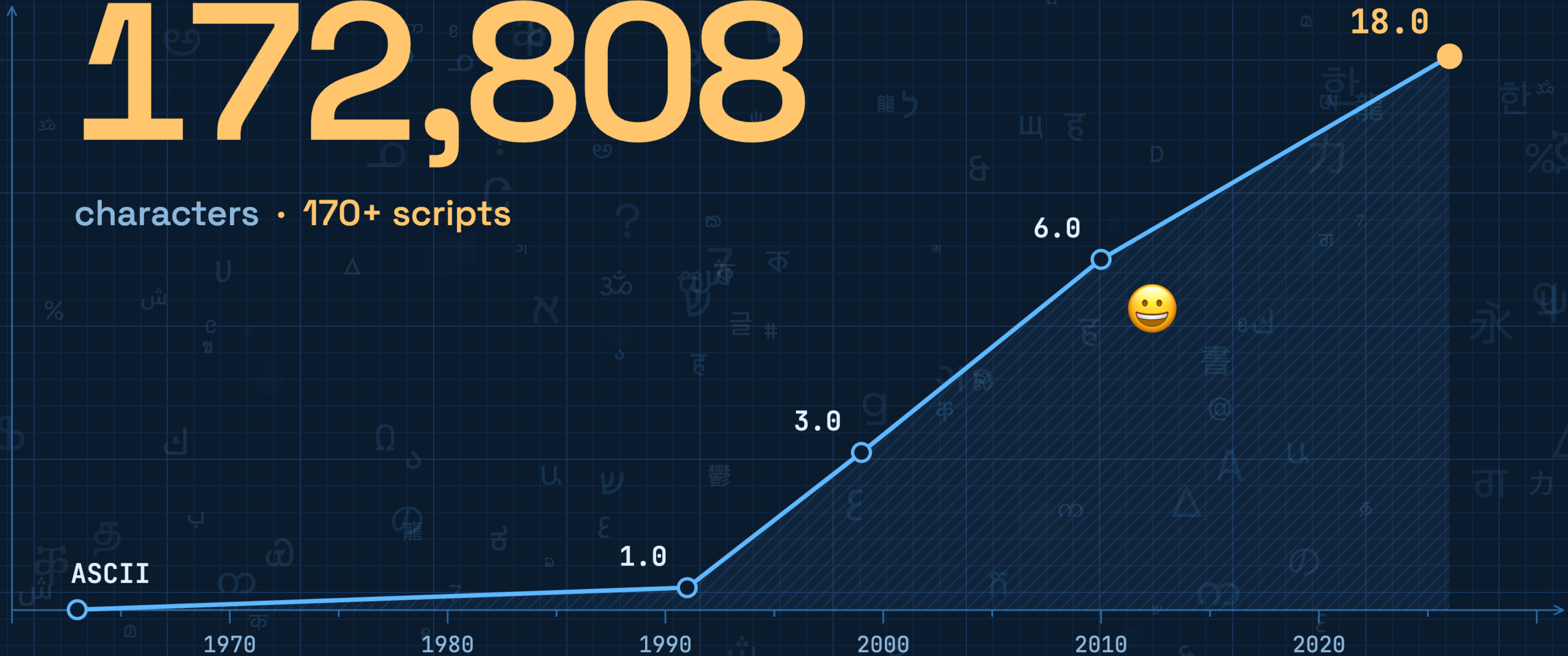
「文字体系ごとに増えるルール」



128 → 172,808

172,808

characters · 170+ scripts



Every script breaks a rule

right-to-left

שלום

joining

ع ع

reordering

memory

क ि

screen

कि

stacking

پ

composition

호 + ㅏ + ㄴ

한

vertical · 縦書き

へ
読
む

右
か
ら
左

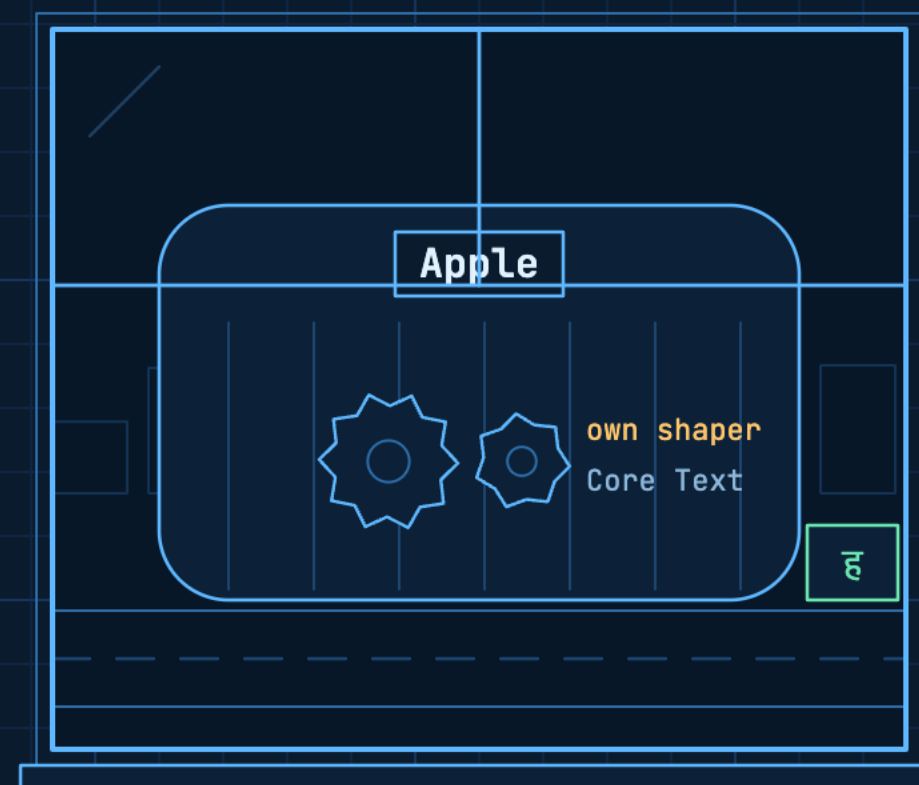
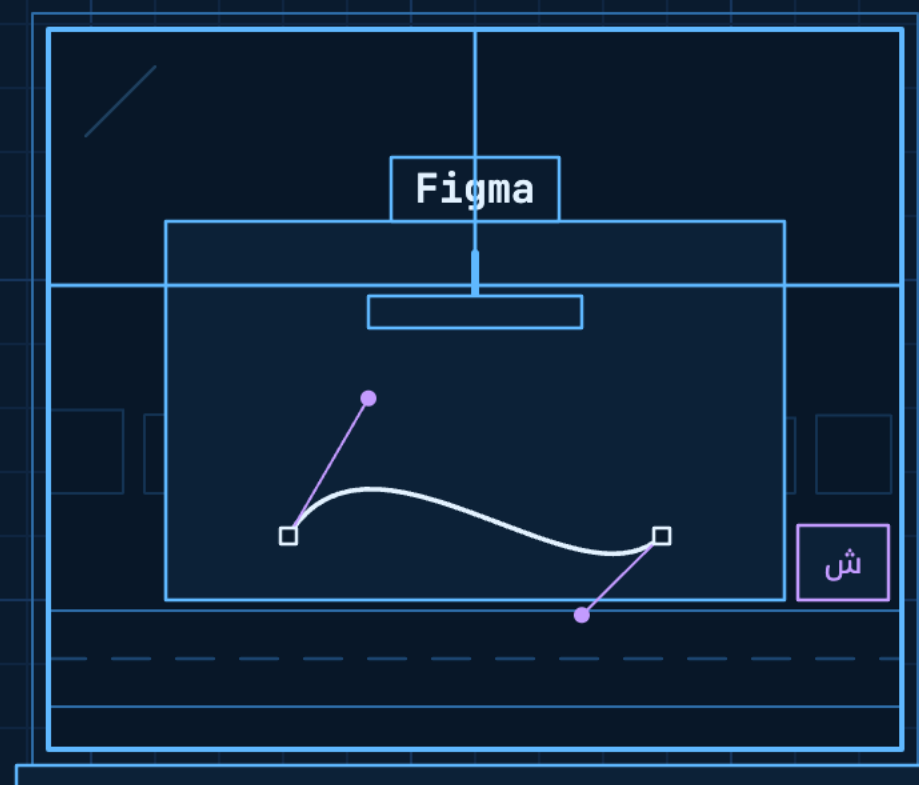
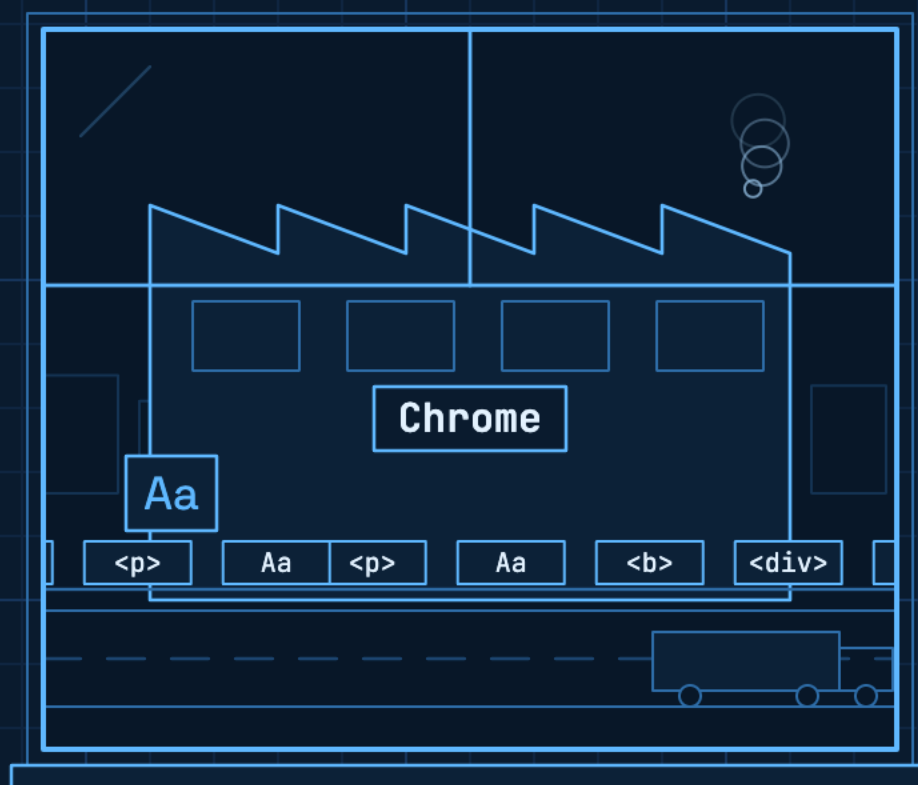
縦
書
き
は

03

hall 03 · other factories

How others do it

「他のエンジンはどうしている？」



Five engines

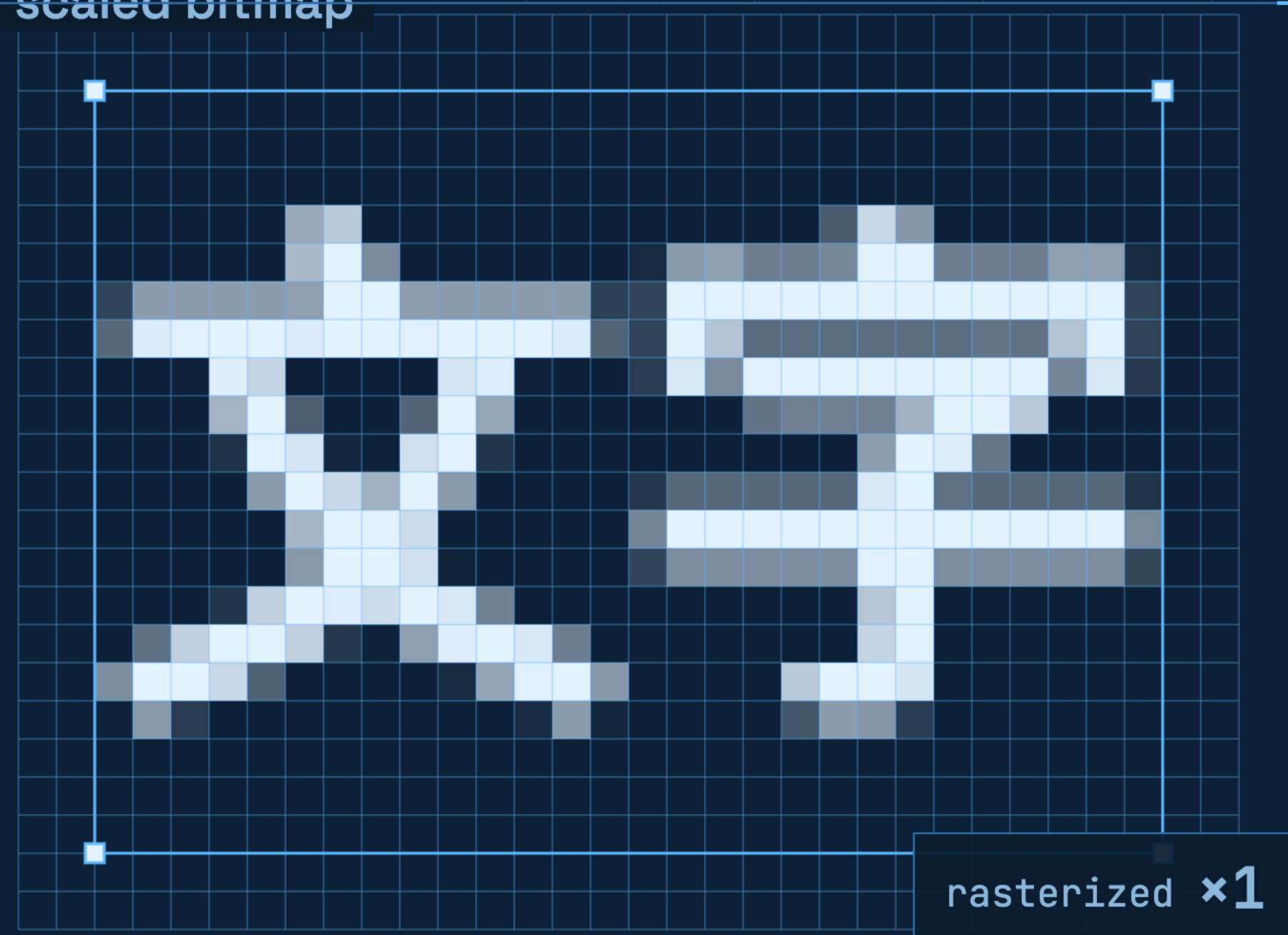
	Chrome	Figma	macOS	Android	◆ Flutter
fonts	system + web	Noto fallback	system cascade	system fallback	FontCollection
shaping	HarfBuzz	HarfBuzz	Core Text	HarfBuzz	HarfBuzz
layout	LayoutNG	own C++	TextKit 2	Minikin	SkParagraph
raster	Skia	WebGL / WebGPU	Core Graphics	Skia	Impeller

HarfBuzz shapes text in 4 of 5

Figma

one canvas, own text engine ≈ Flutter

scaled bitmap



re-rendered



zoom

20×

auto

hall 03
others

01

02

03

04

05

end

14 / 26

04

hall 04 · our specs

Flutter's trade-off

「Flutterのトレードオフ」

flutter · spec

missing



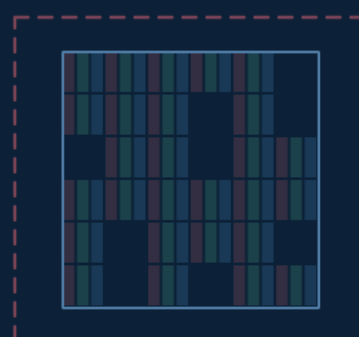
vertical press



ruby press



hyphenator



LCD painter



only here



ios android web

same parts



shaders



text  text

widgets in text



Not in Flutter

workarounds

縦書き

vertical text



かん じ
漢 字

ruby / furigana



hyphen-
ation

hyphenation



نص جميل
نص جميل

kashida justify



🔍 text 1/3

the **text** on a page
is real DOM **text**:
find · translate
every **text** node

find & translate (web)



65 E5 67 2C 8A 9E

fonts on demand (web)



Only in Flutter

same everywhere

paint

widgets in text

every glyph

3D

Text

نص · 文字 · ข้อความ

rotateX -1° rotateY 30°

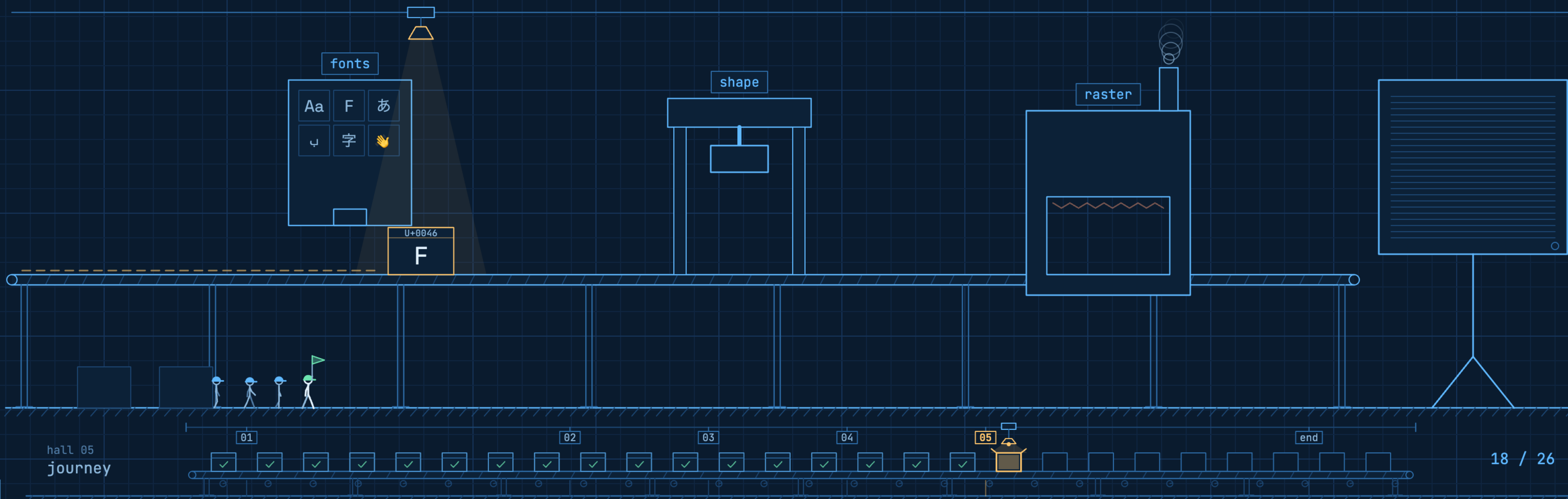
edit text

05

hall 05 · follow one crate

One word's journey

「ひとつの単語の旅」



Text('Flutter')

```
Text(  
  'Flutter',  
  style: TextStyle(  
    fontSize: 72,  
  ),  
)
```

Text

widgets

build()

RichText

widgets

createRenderObject()

RenderParagraph

rendering

owns

TextPainter

painting

builds

ui.Paragraph

dart:ui

Flutter

constraints ↓ $w \leq 400$
size ↑ 226 × 92

widget

Flutter

Type

مرحبا



Find the glyphs

glyphs

Flutter Type مرحبا

F l u t t e r

#9 #41 #50 #49 #49 #34 #47



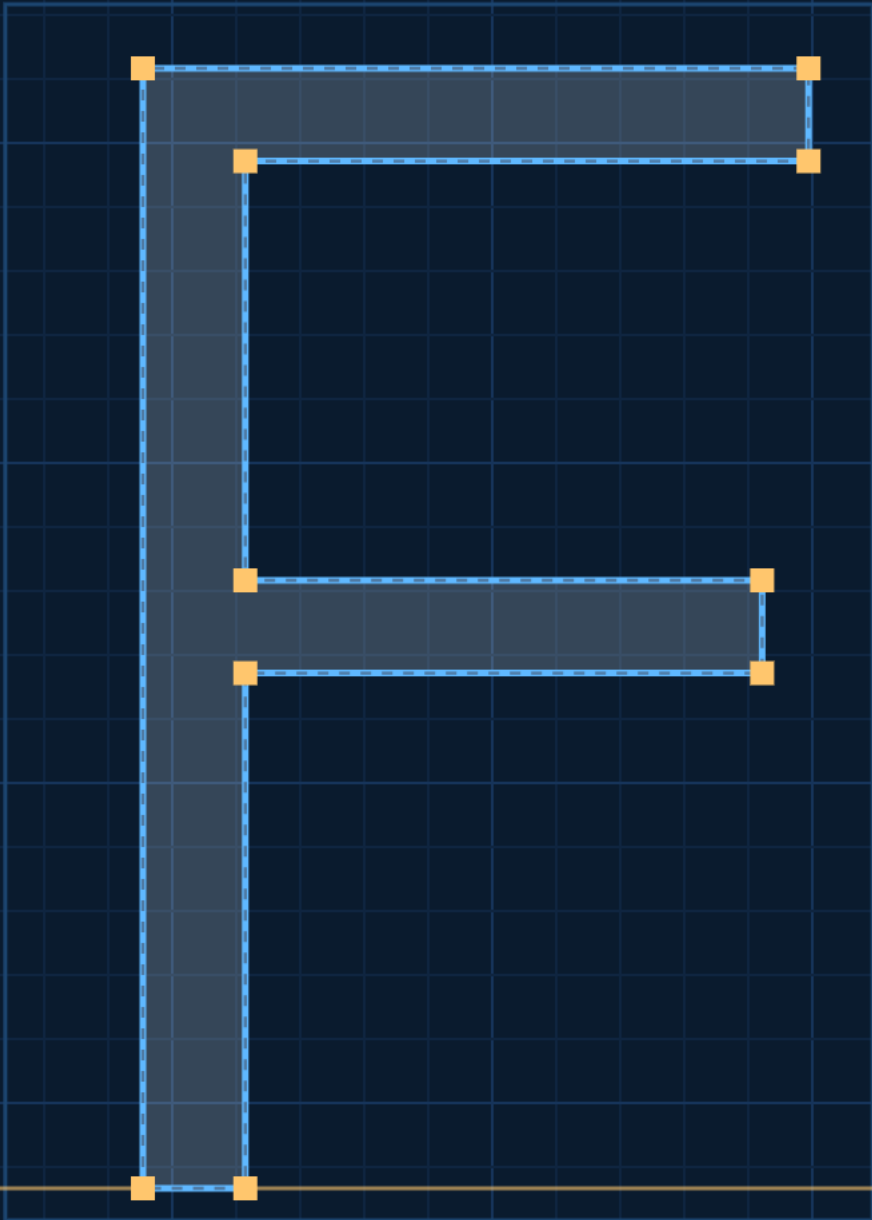
U+0046

fontFamily
Space Grotesk ✓

fontFamilyFallback
Noto Kufi Arabic

platform
system fonts

cmap
#9



Shape

shape

Flutter

Type

مرحبا



F U+0046

l U+006C

u U+0075

t U+0074

t U+0074

e U+0065

r U+0072

HarfBuzz
hb_shape()

Flutter

hmtx 3280

#9 #41 #50 #49 #49 #34 #47

Flutter

shaped 3138

542 236 611 391 391 598 369

kern

hall 05

journey

01



02



03



04



05



end

21 / 26

Lay out

layout

Flutter

Type

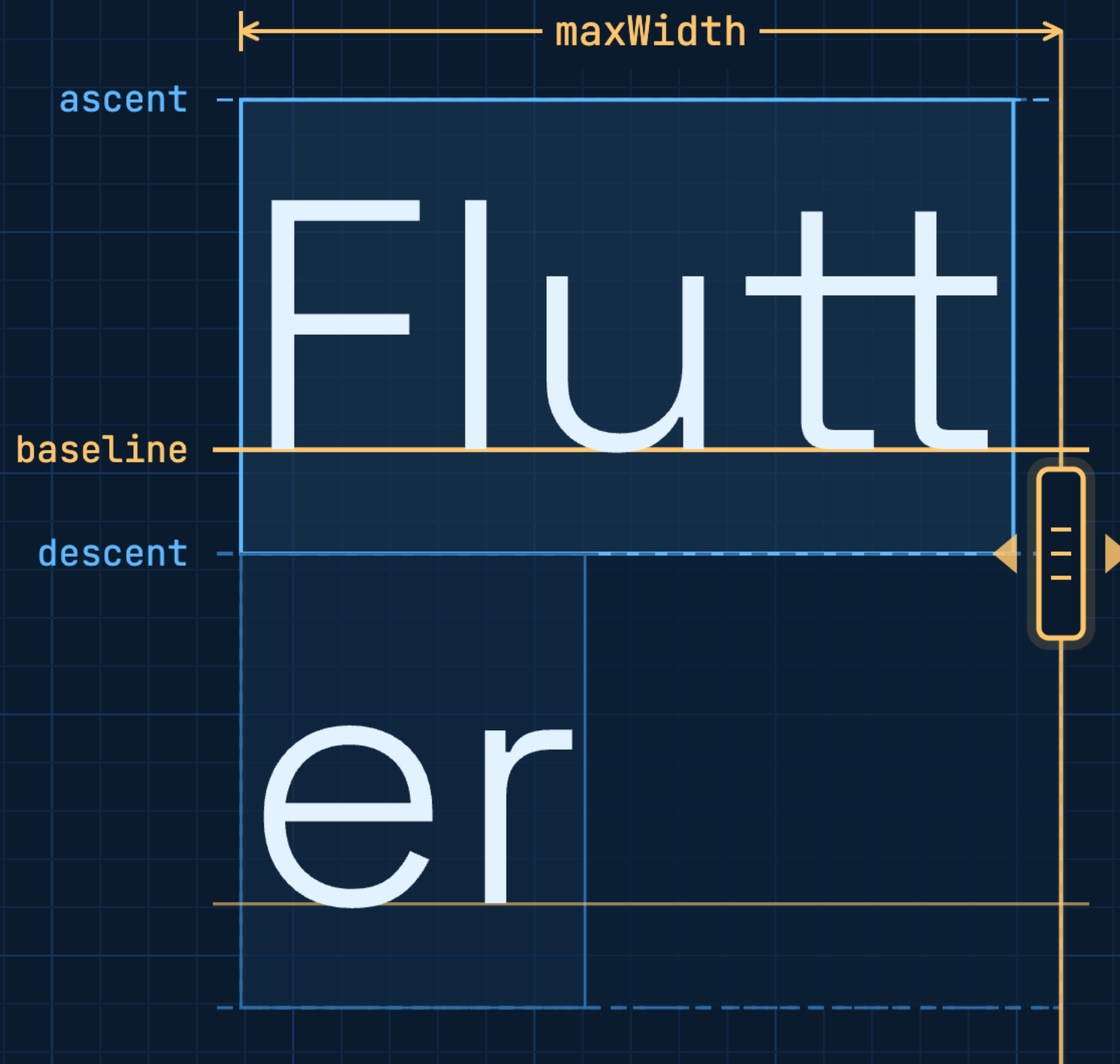
مرحبا



left

center

right



width

408

height

452

lines

2

shape
×1

wrap
×142

Rasterize

raster

Flutter

Type

مرحبا



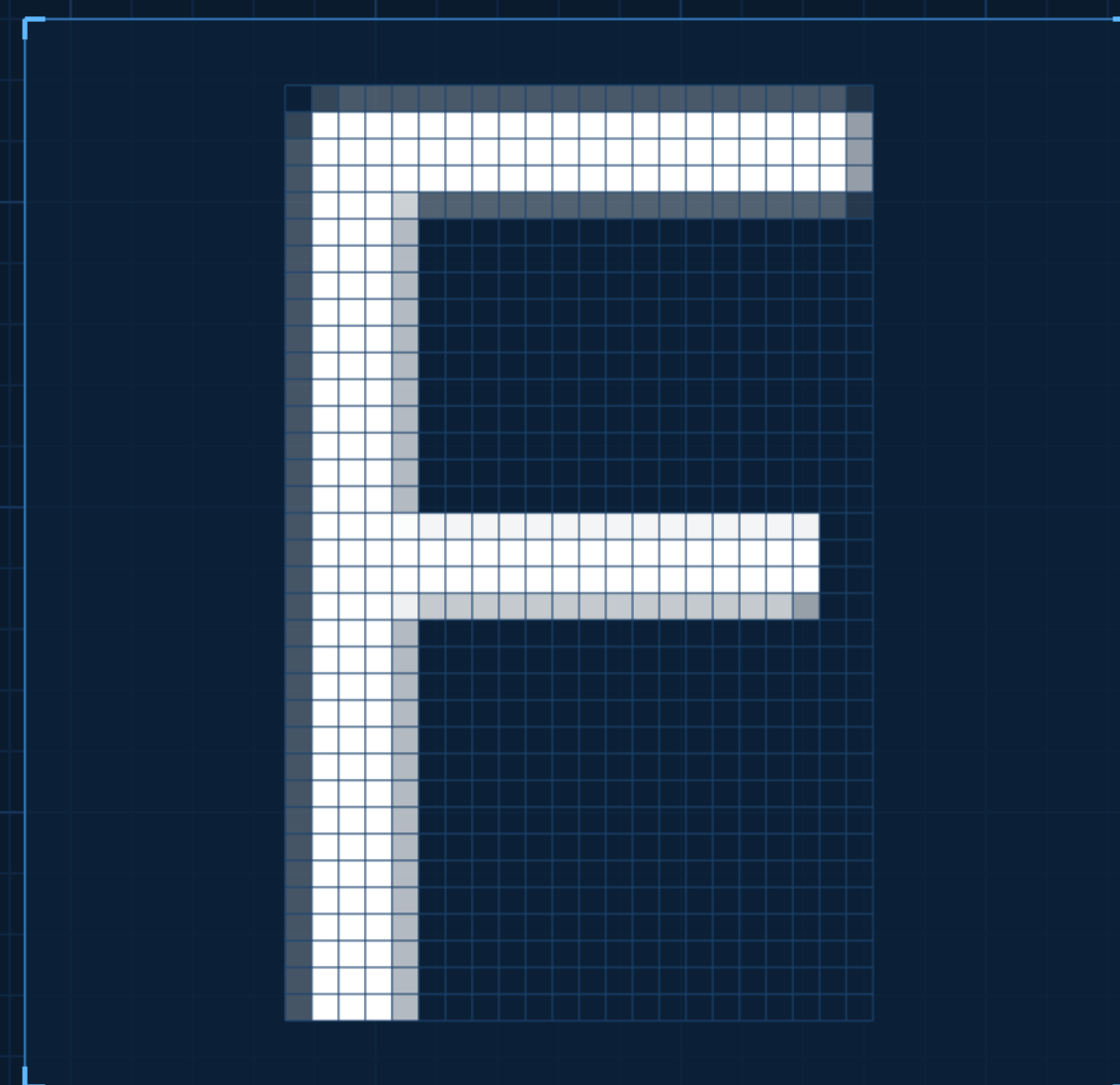
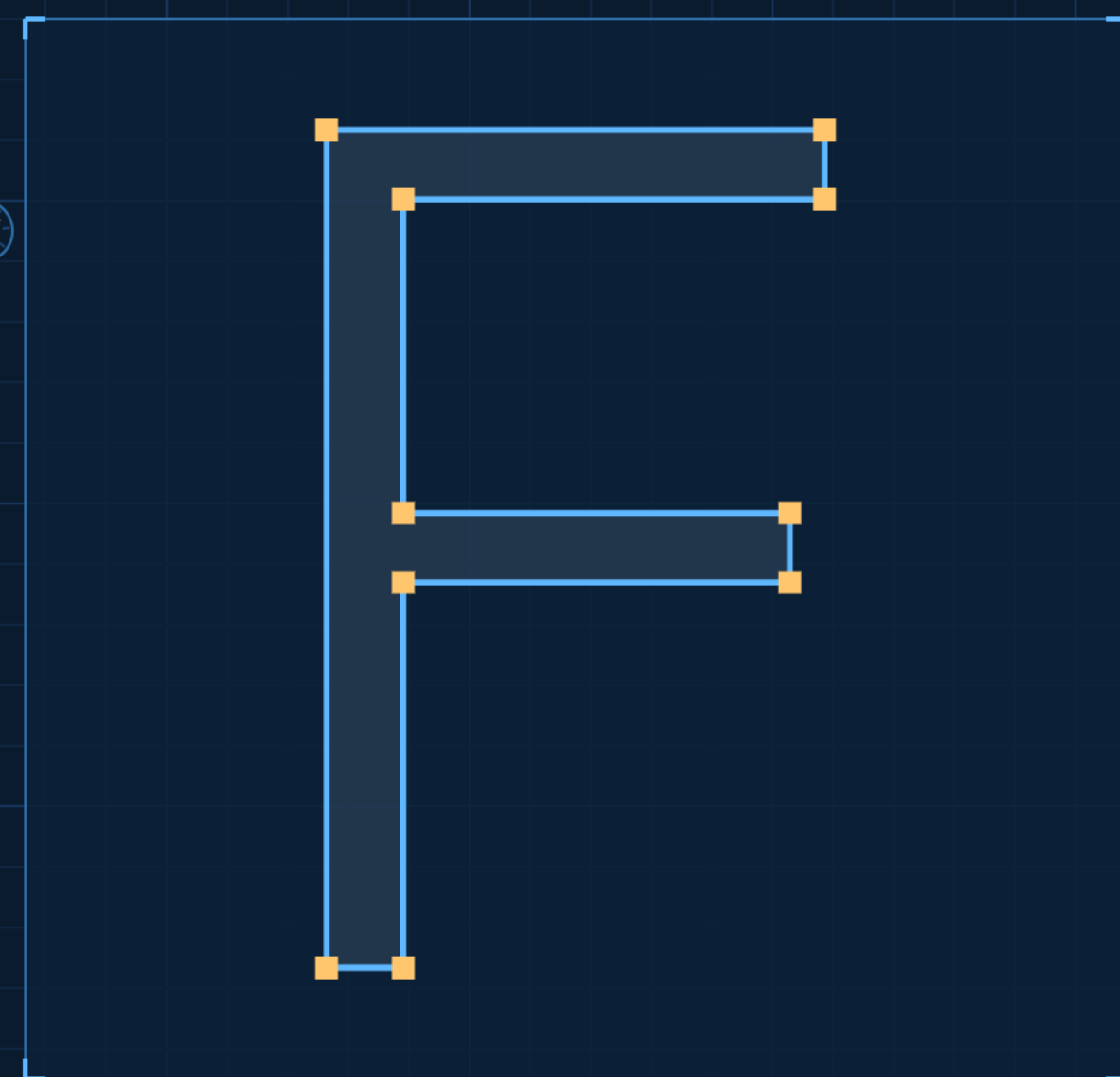
size

48 px

outline

pixels

atlas



glyph #9

22 × 35 px

7 tiles · 128 × 128

hall 05
journey

01

02

03

04

05

end

23 / 26

Draw

draw

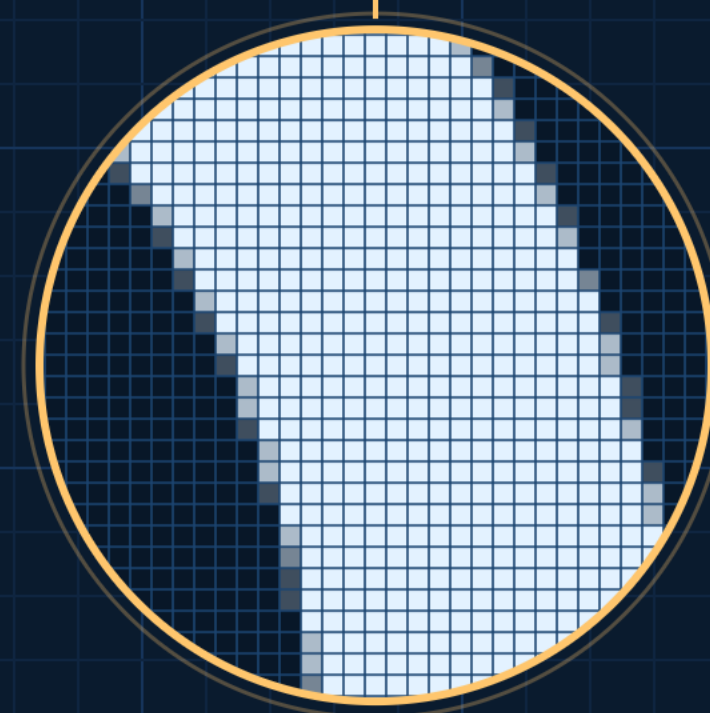
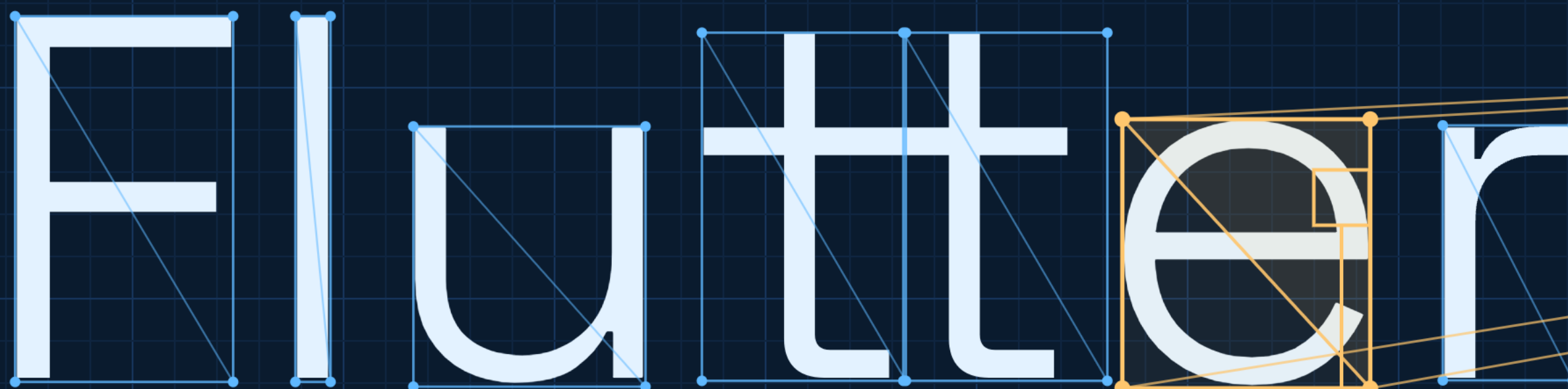
Flutter

Type

مربعاً



glyph atlas

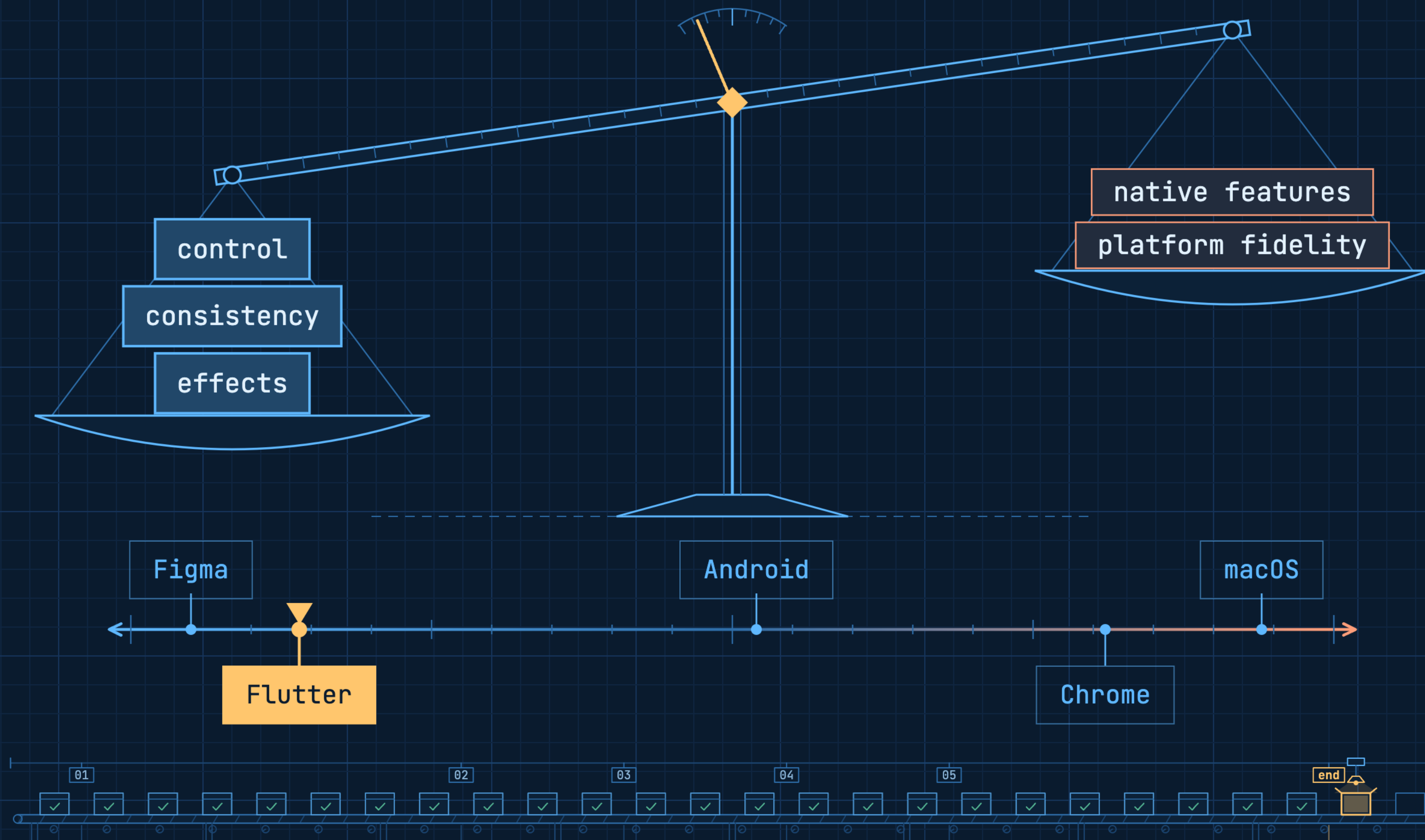


7 glyphs

14 triangles

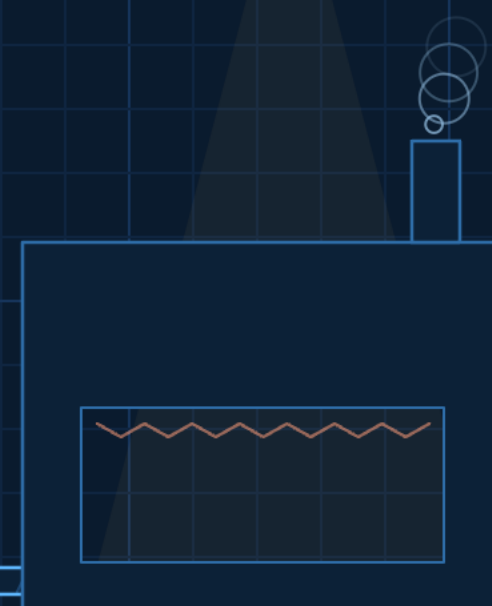
1 draw call

The trade-off



「ありがとうございました」

actions? ・ 質問は? ・ ありがとうございました ・ thank you ・ questi



あ り が と う



exit
outro

01

02

03

04

05

end